

GENIFY

LECTURE EXTRA

29-September-2025

TOPICS COVERED:

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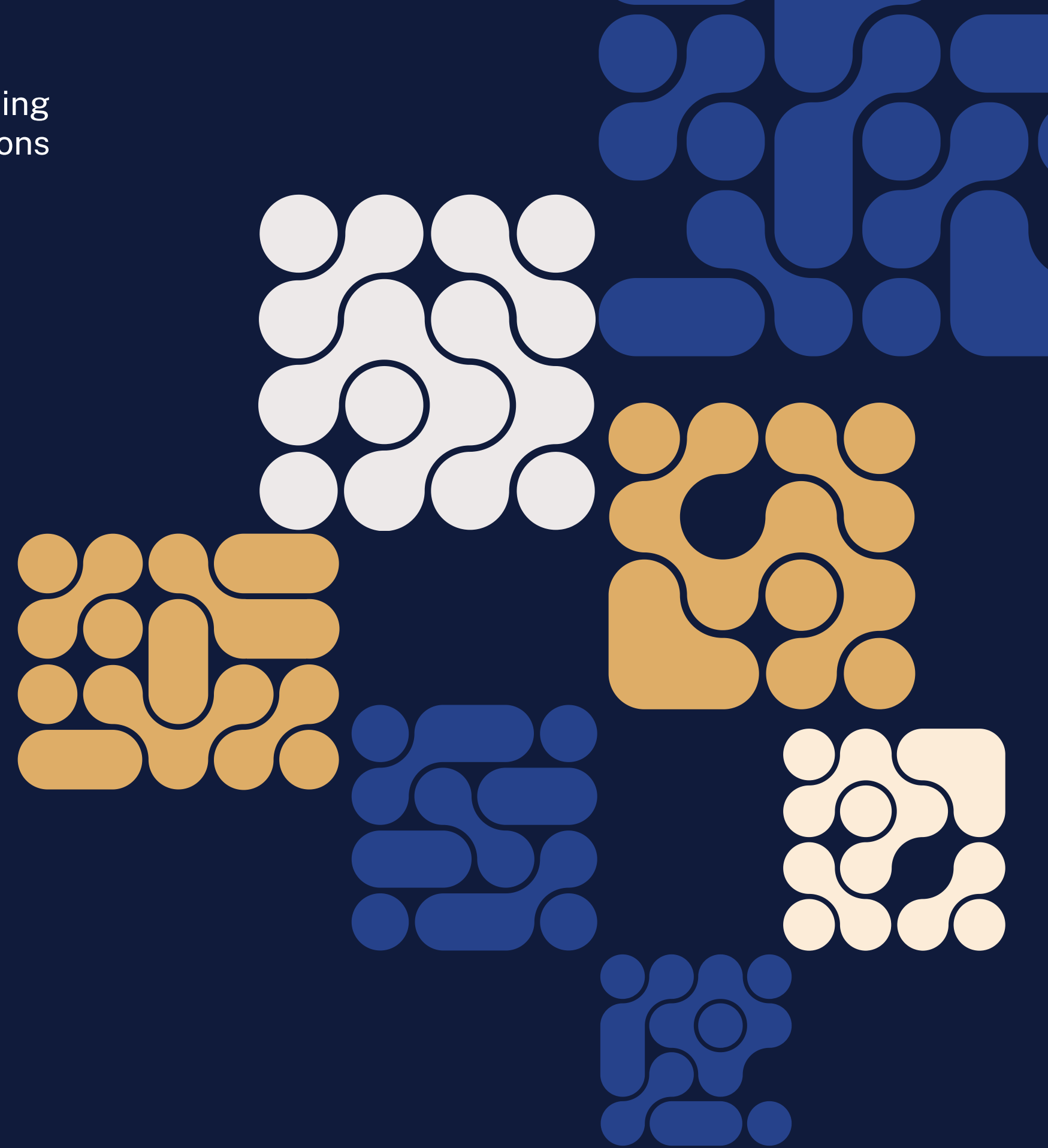
- A.I AGENTS
- A.I AGENTS FRAMEWORK

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A.I AGENTS

Q: What are Agents in A.I ?

- An A.I Agent is a software program that can interact with its environment, collect data and use that data to perform self-directed tasks that meet predetermined goals.
- Humans set goals, but an AI agent independently chooses the best actions it needs to perform to achieve those goals.
- For example, consider a customer support AI agent that wants to resolve customer queries. The agent will automatically ask the customer different questions, look up information in internal documents and respond with a solution. Based on the customer responses, it determines if it can resolve the query itself or pass it on to a human.

Key Features of A.I Agents

- **Autonomous:** Act without constant human input and decide next steps from past data like a bookstore bot flags missing invoices.
- **Goal-driven:** Optimize for defined objectives like a logistics AI balancing speed, cost and fuel use.
- **Perceptive:** Gather info from sensors, inputs or APIs like a cybersecurity agent tracking new threats.
- **Adaptable:** Adjust strategies when situations change.
- **Collaborative:** Work with humans or other agents toward shared goals like healthcare agents coordinating with patients and doctor.

How do AI Agents Work?

1. Persona: Each agent is given a clearly defined role, personality and communication style along with specific instructions and descriptions of the tools it can use. A well-crafted persona ensures the agent behaves consistently and appropriately for its role, while also evolving as it gains experience and engages with users or other systems.

2. Memory: Agents typically have multiple types of memory:

- Short-term memory for the current interaction
- Long-term memory for storing historical data and conversations
- Episodic memory for recalling specific past events
- Consensus memory for sharing knowledge among multiple agents

Memory enables an agent to keep context, learn from experience and adapt its behaviour over time.

How do AI Agents Work?

3. Tools: These are functions or external resources the agent can use to access information, process data, control devices or connect with other systems. Tools may involve physical interfaces, graphical UIs or programmatic APIs. Agents also learn how and when to use these tools effectively, based on their capabilities and context.

4. Model: Agents use large language model (LLM) which serves as the agent's "brain". The LLM interprets instructions, reasons about solutions, generates language and orchestrates other components including memory retrieval and tools to use to carry out tasks.

Architecture of AI Agents

There are four main components

- **Profiling Module:** This module helps the agent understand its role and purpose. It gathers information from the environment to form perceptions. For example: A self-driving car uses sensors and cameras to detect obstacles.
- **Memory Module:** The memory module enables the agent to store and retrieve past experiences. This helps the agent learn from prior actions and improve over time. For example: A chatbot remembers past conversations to give better responses.
- **Planning Module:** This module is responsible for decision-making. It evaluates situations, weighs alternatives and selects the most effective course of action. For example: A chess-playing AI plans its moves based on future possibilities.
- **Action Module:** The action module executes the decisions made by the planning module in the real world. It translates decisions into real-world actions. For example: A robot vacuum moves to clean a designated area after detecting dirt.

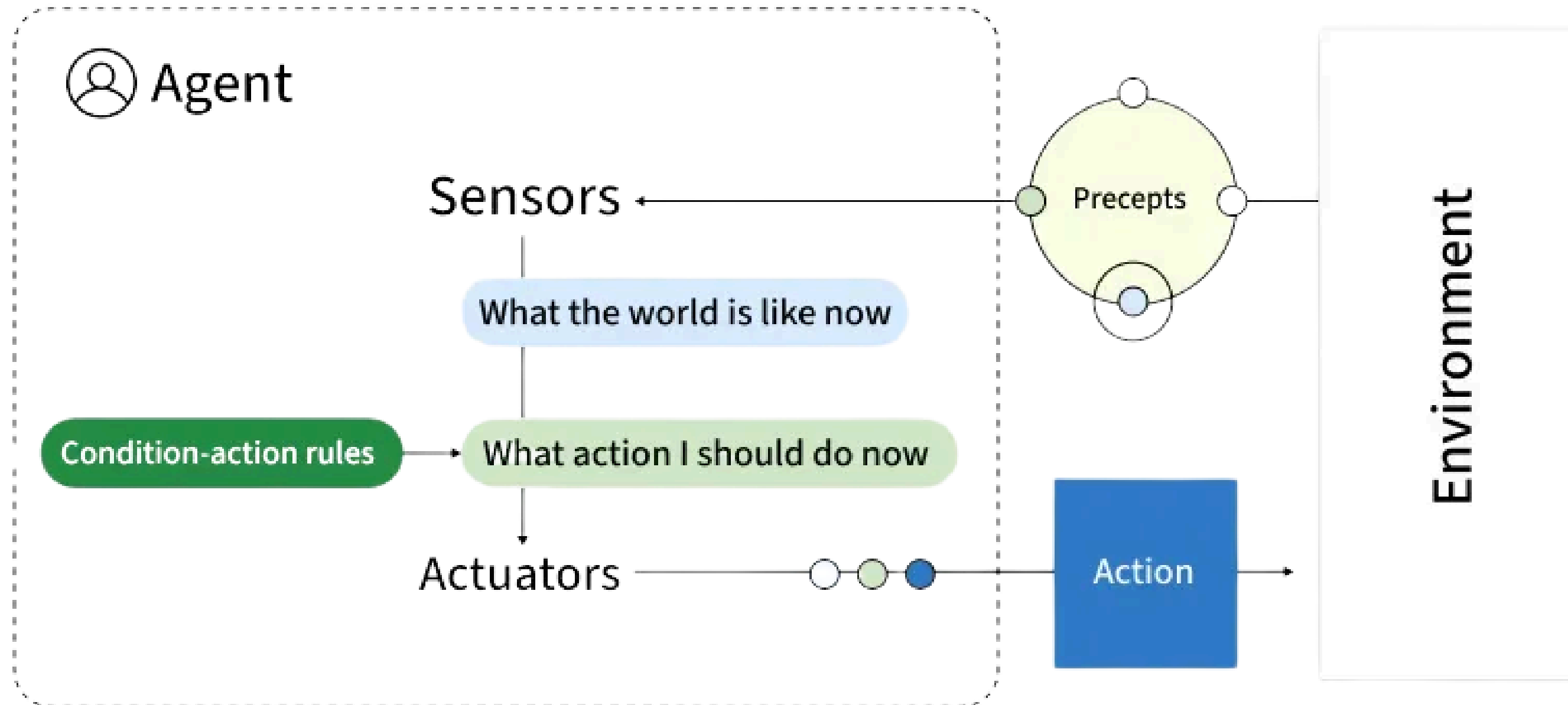
AI Agent Classification

An agent is a system designed to perceive its environment, make decisions and take actions to achieve specific goals. Agents operate autonomously, without direct human control and can be classified based on their behavior, environment and number of interacting agents.

- **Reactive Agents:** Respond to immediate environmental stimuli without foresight or planning.
- **Proactive Agents:** Anticipate future states and plan actions to achieve long-term goals.
- **Single-Agent Systems:** One agent solves a problem independently.
- **Multi-Agent Systems:** Multiple agents interact, coordinate or compete to achieve goals; may be homogeneous (similar roles) or heterogeneous (diverse roles).
- **Rational Agents:** Choose actions to maximize expected outcomes using both current and historical information.

Types of Agents

1. Simple Reflex Agents



Types of Agents

1. Simple Reflex Agents

Simple Reflex Agents act based solely on current perceptions using condition-action rules. These agents respond directly to stimuli without considering past experiences or potential future states. They operate on basic "if-then" logic: if a specific condition is detected, execute a corresponding action.

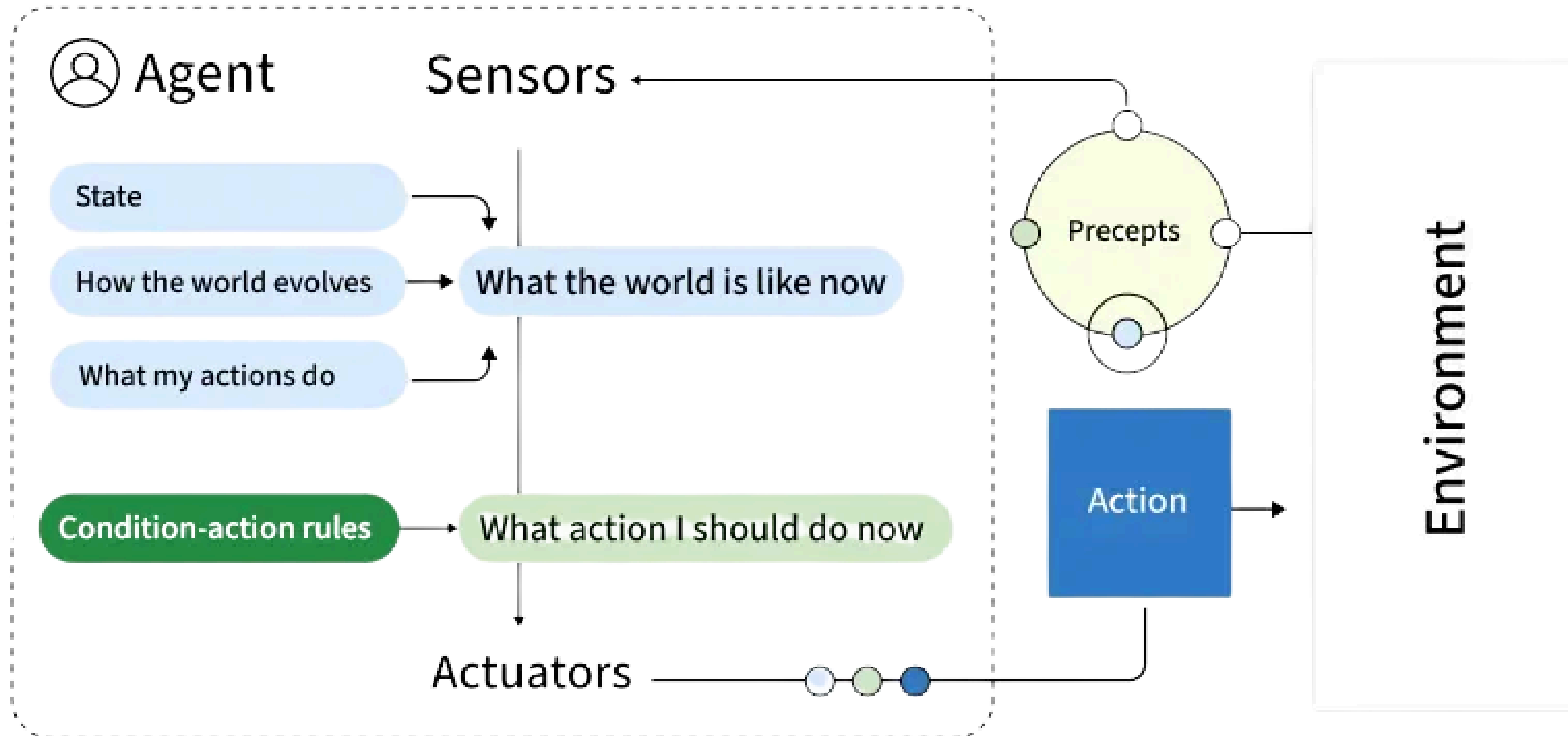
Key Features:

- No memory of past states
- No model of how the world works
- Purely reactive behavior
- Function best in fully observable environments

For Example, Traffic light control systems that change signals based on fixed timing.

Types of Agents

2. Model-Based Reflex Agents



Types of Agents

2. Model-Based Reflex Agents

Model based Reflex Agents maintain an internal representation of the world, allowing them to track aspects of the environment they cannot directly observe. This internal model helps them make more informed decisions by considering how the world evolves and how their actions affect it.

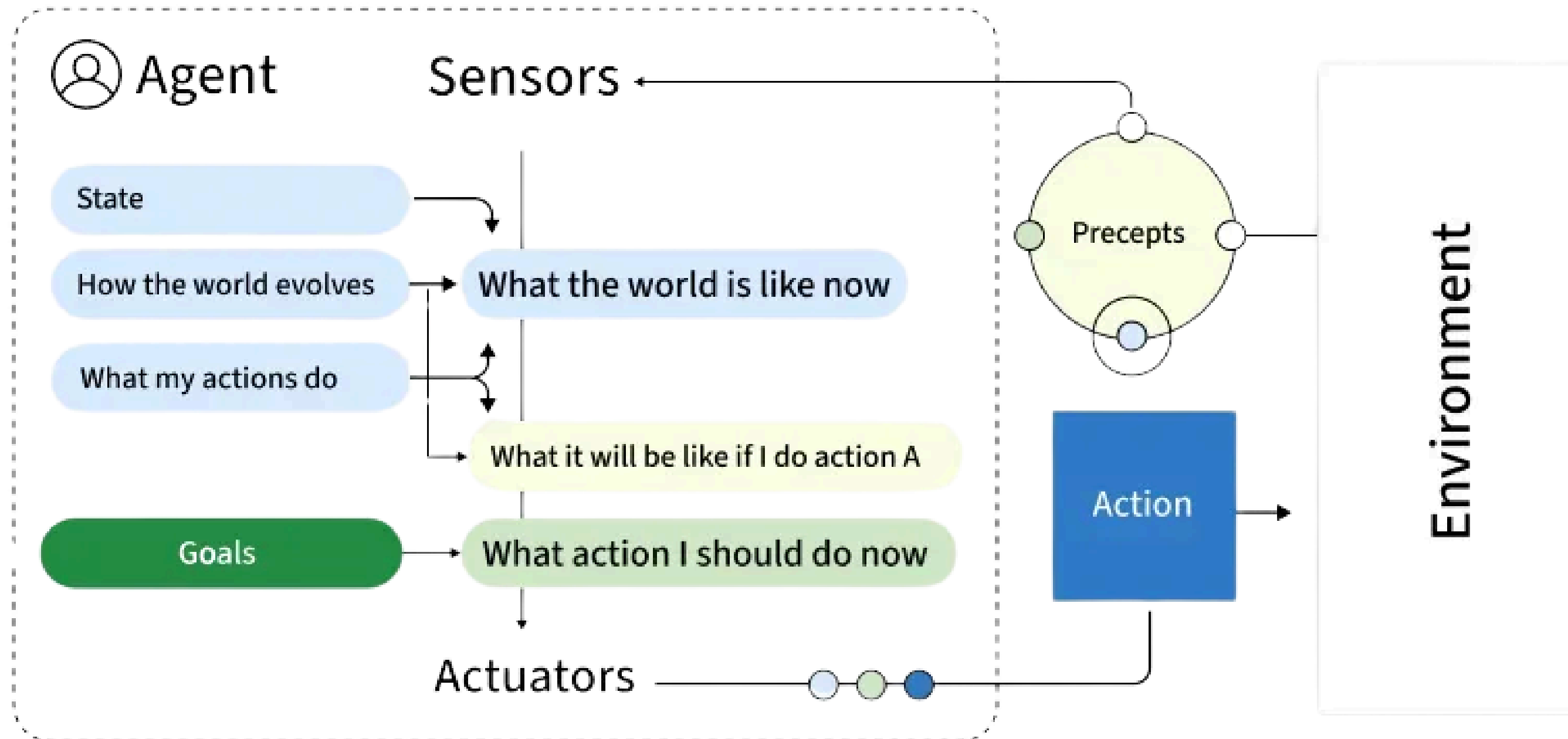
Key Features:

- Track the world's state over time
- Infer unobserved aspects of current states
- Function effectively in partially observable environments
- Still primarily reactive, but with contextual awareness

For example, Robot vacuum cleaners that map rooms and tracks cleaned areas.

Types of Agents

3. Goal-Based Agents



Types of Agents

3. Goal-Based Agents

Goal based Agents plan their actions with a specific objective in mind. Unlike reflex agents that respond to immediate stimuli, goal-based agents evaluate how different action sequences might lead toward their defined goal, selecting the path that appears most promising.

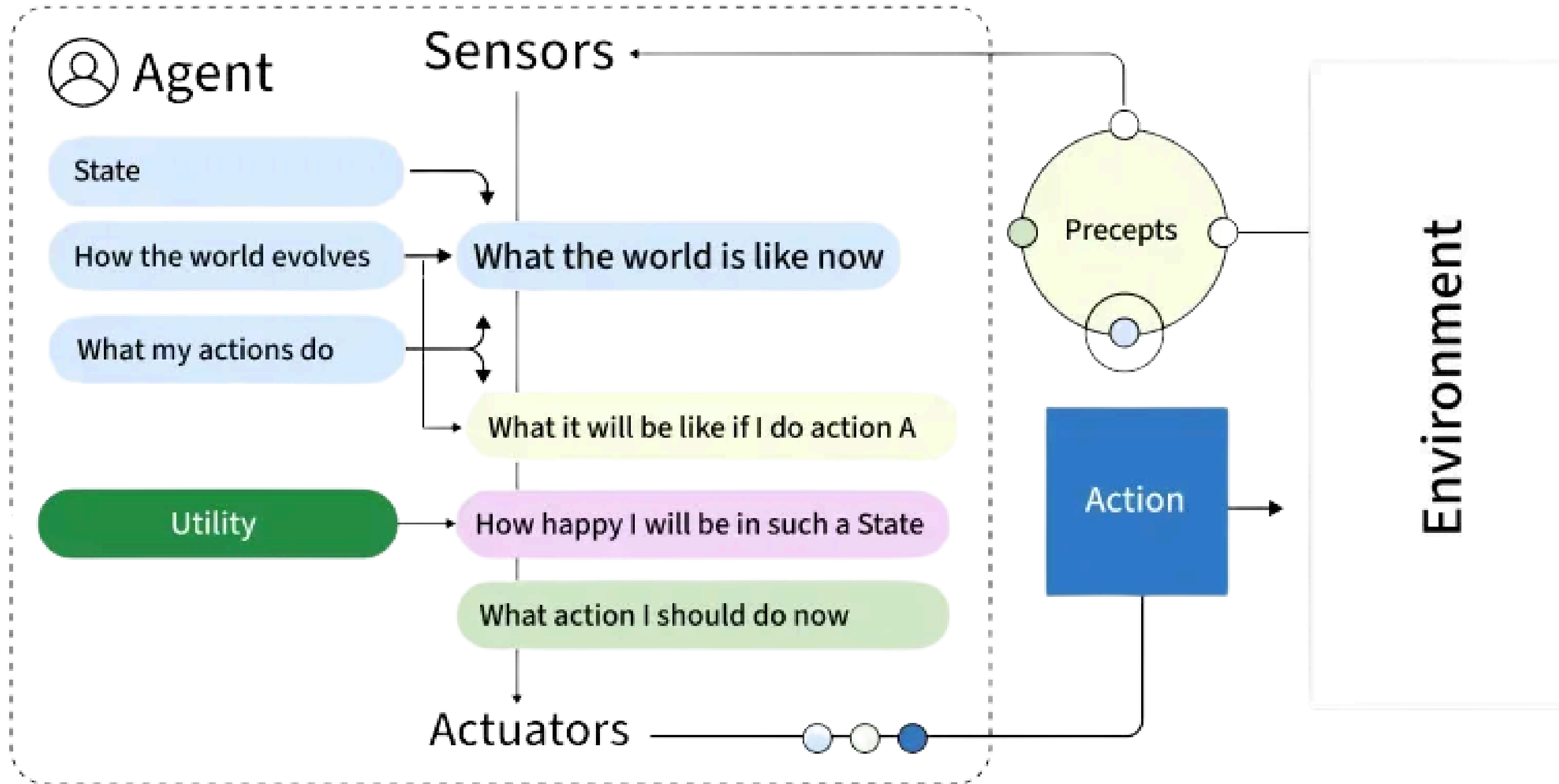
Key Features:

- Employ search and planning mechanisms
- Evaluate actions based on their contribution toward goal achievement
- Consider future states and outcomes
- May explore multiple possible routes to a goal

For example, Logistics routing agents that find optimal delivery routes based on factors like distance and time. They continually adjust to reach the most efficient route.

Types of Agents

4. Utility-Based Agents



Types of Agents

4. Utility-Based Agents

Utility based Agents extend goal-based thinking by evaluating actions based on how well they maximize a utility function—essentially a measure of "happiness" or "satisfaction." This approach allows them to make nuanced trade-offs between competing goals or uncertain outcomes.

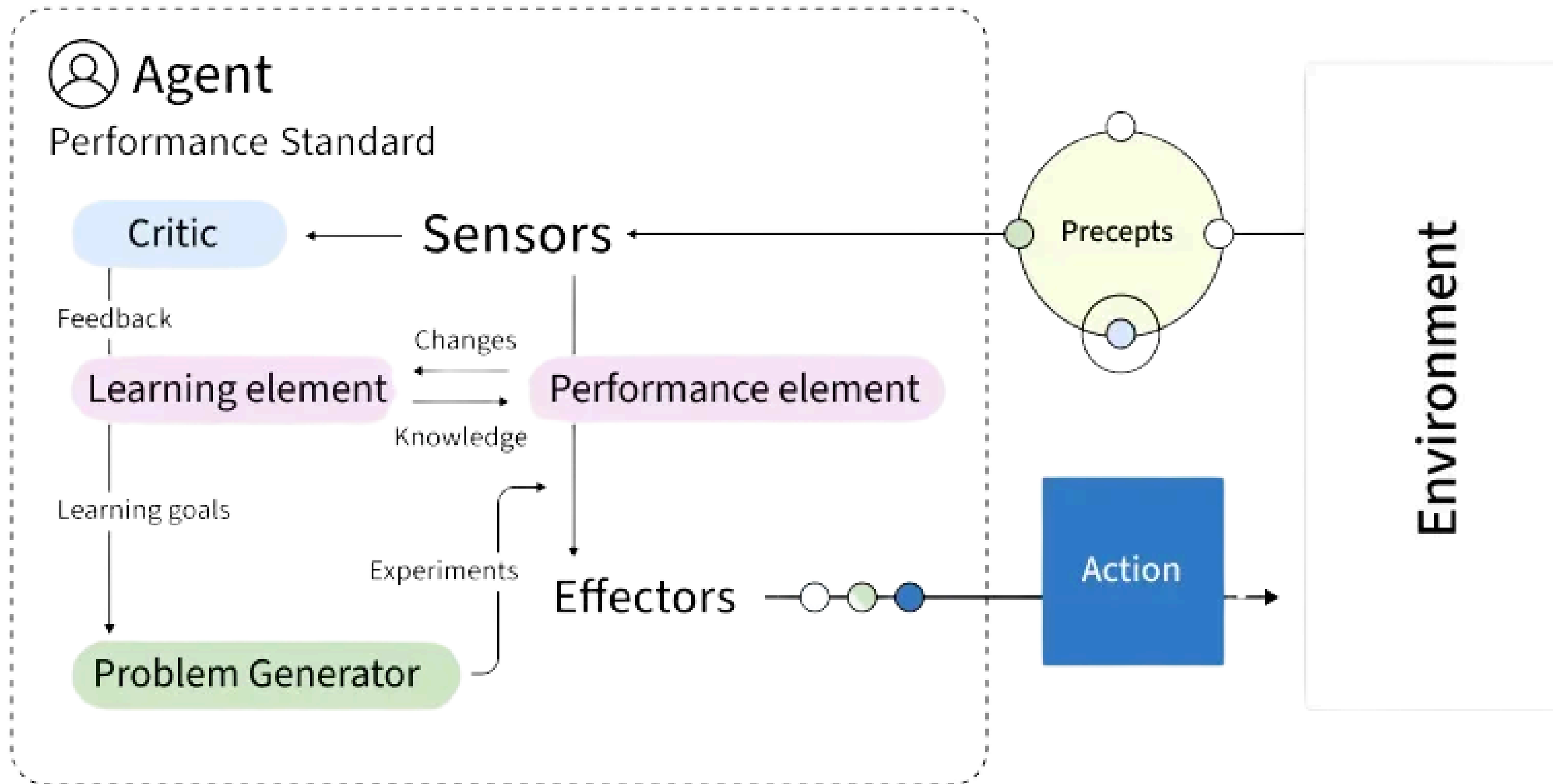
Key Features:

- Balance multiple, sometimes conflicting objectives
- Handle probabilistic and uncertain environments
- Evaluate actions based on expected utility
- Make rational decisions under constraints

For example, Financial portfolio management agents that evaluate investments based on factors like risk, return and diversification operate by choosing options that provide the most value.

Types of Agents

5. Learning Agents



Types of Agents

5. Learning Agents

Learning Agents improve their performance over time based on experience. They modify their behavior by observing the consequences of their actions, adjusting their internal models and decision-making approaches to achieve better outcomes in future interactions.

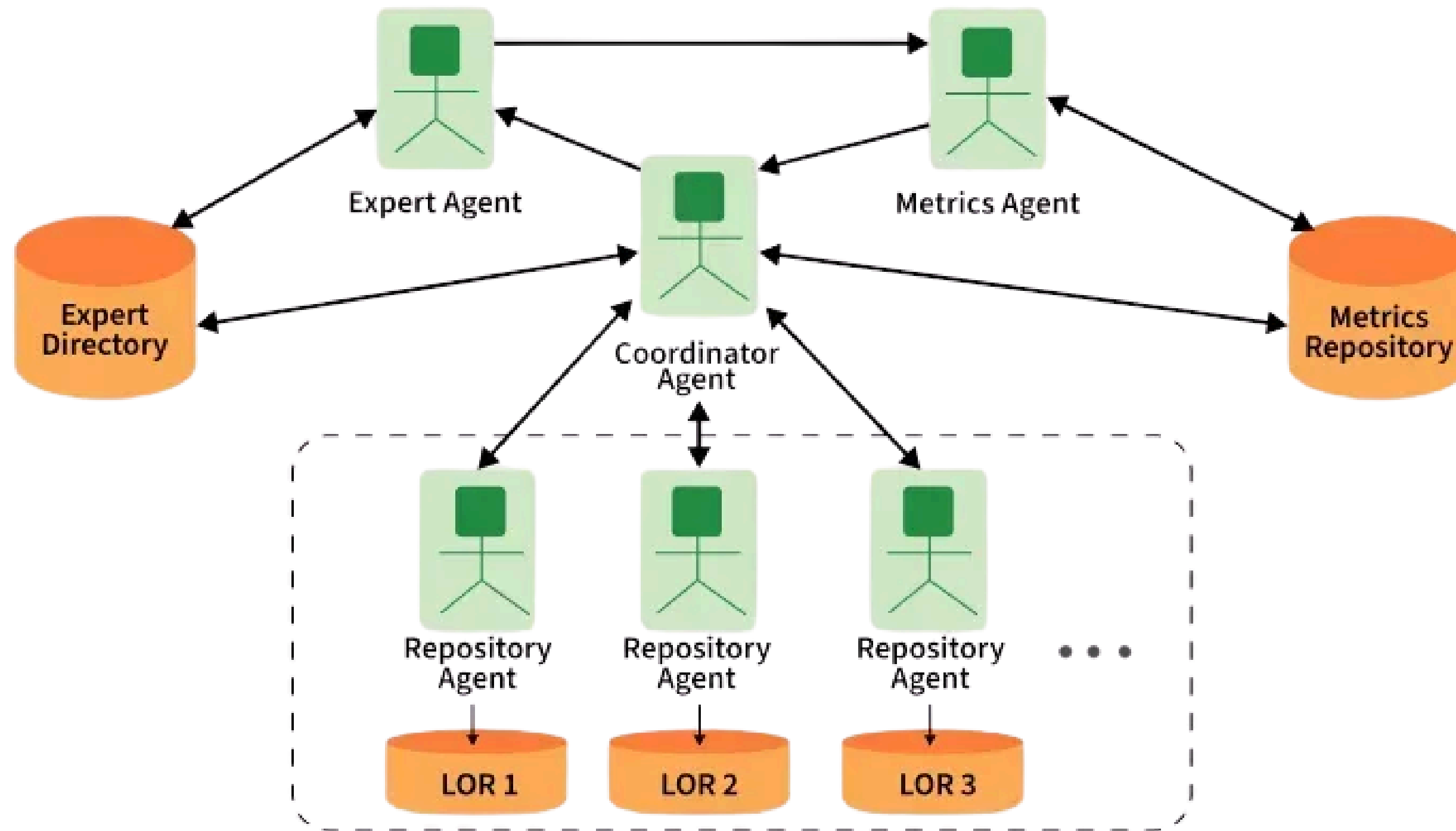
Key Features:

- Adapt to changing environments
- Improve performance with experience
- Contain both a performance element and a learning element
- Generate new knowledge rather than simply applying existing rules

For example, Customer service chatbots can improve response accuracy over time by learning from previous interactions and adapting to user needs.

Types of Agents

6. Multi-Agent Systems (MAS)



Types of Agents

6. Multi-Agent Systems (MAS)

Multi-Agent Systems consist of multiple autonomous agents that interact with each other within an environment. These agents may cooperate toward common goals, compete for resources or exhibit a mix of cooperative and competitive behaviors. Types of multi-agent systems:

- **Cooperative MAS:** Agents work together toward shared objectives.
- **Competitive MAS:** Agents pursue individual goals that may conflict.
- **Mixed MAS:** Agents cooperate in some scenarios and compete in others.

Key Features:

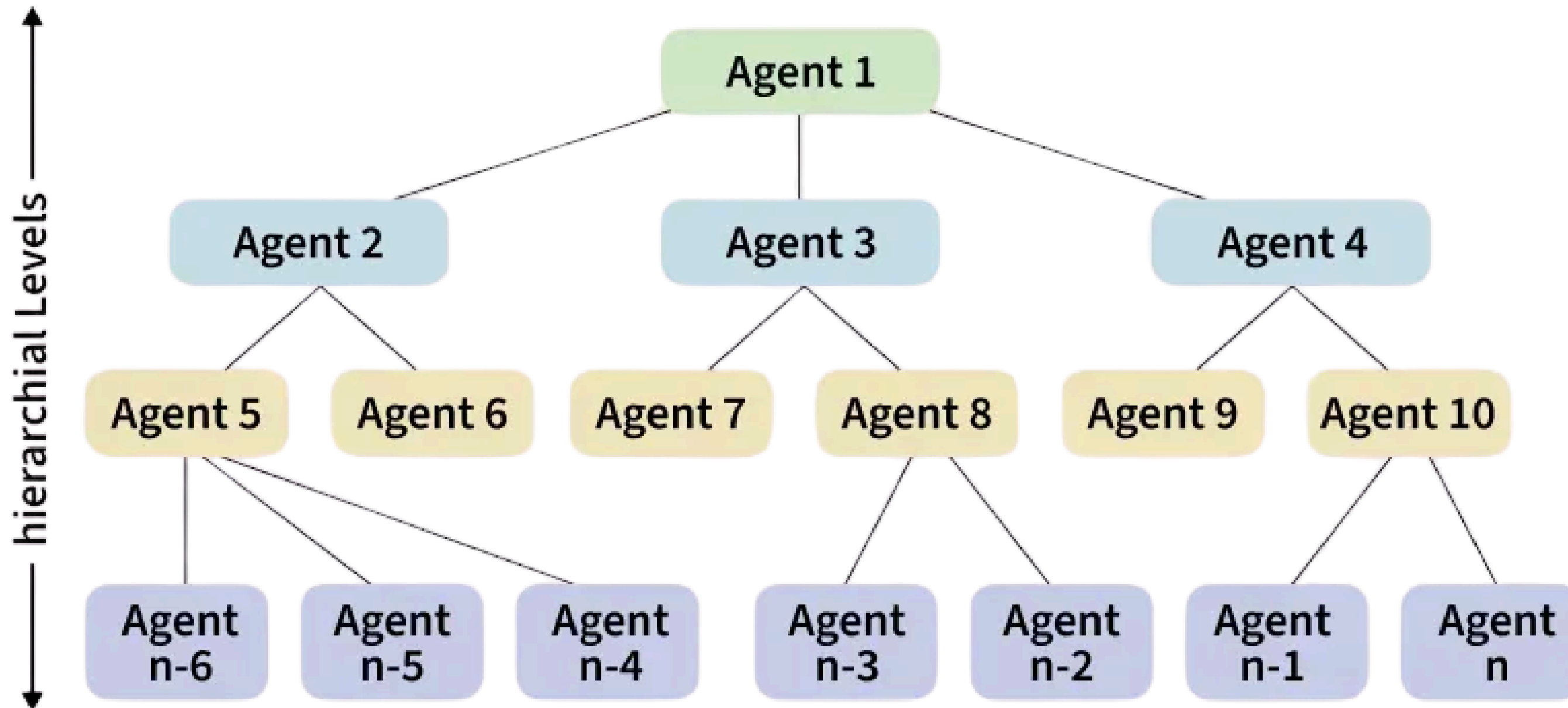
- Agents act independently and control their own state.
- Agents align, collaborate or compete to reach goals.
- The system remains resilient if individual agents fail.
- Decisions are distributed; there's no single controller.

For example, a warehouse robot might use:

- Model-based reflexes for navigation
- Goal-based planning for task sequencing
- Utility-based decision-making for prioritizing tasks
- Learning capabilities for route optimization

Types of Agents

7. Hierarchical agents



Types of Agents

7. Hierarchical agents

Hierarchical agents organize decision-making across multiple levels, with high-level agents making strategic decisions and delegating specific tasks to lower-level agents. This structure mirrors many human organizations and allows for managing problems at appropriate levels of abstraction.

Key Features:

- Division of responsibilities across multiple levels
- Abstract decision-making at higher levels
- Detailed execution at lower levels
- Simplified information flow (higher levels receive summarized data)

For example, Drone delivery systems in which fleet management is done at top level and individual navigation at lower level.

Use Cases of AI Agents

Agents are used in a wide range of applications in artificial intelligence, including:

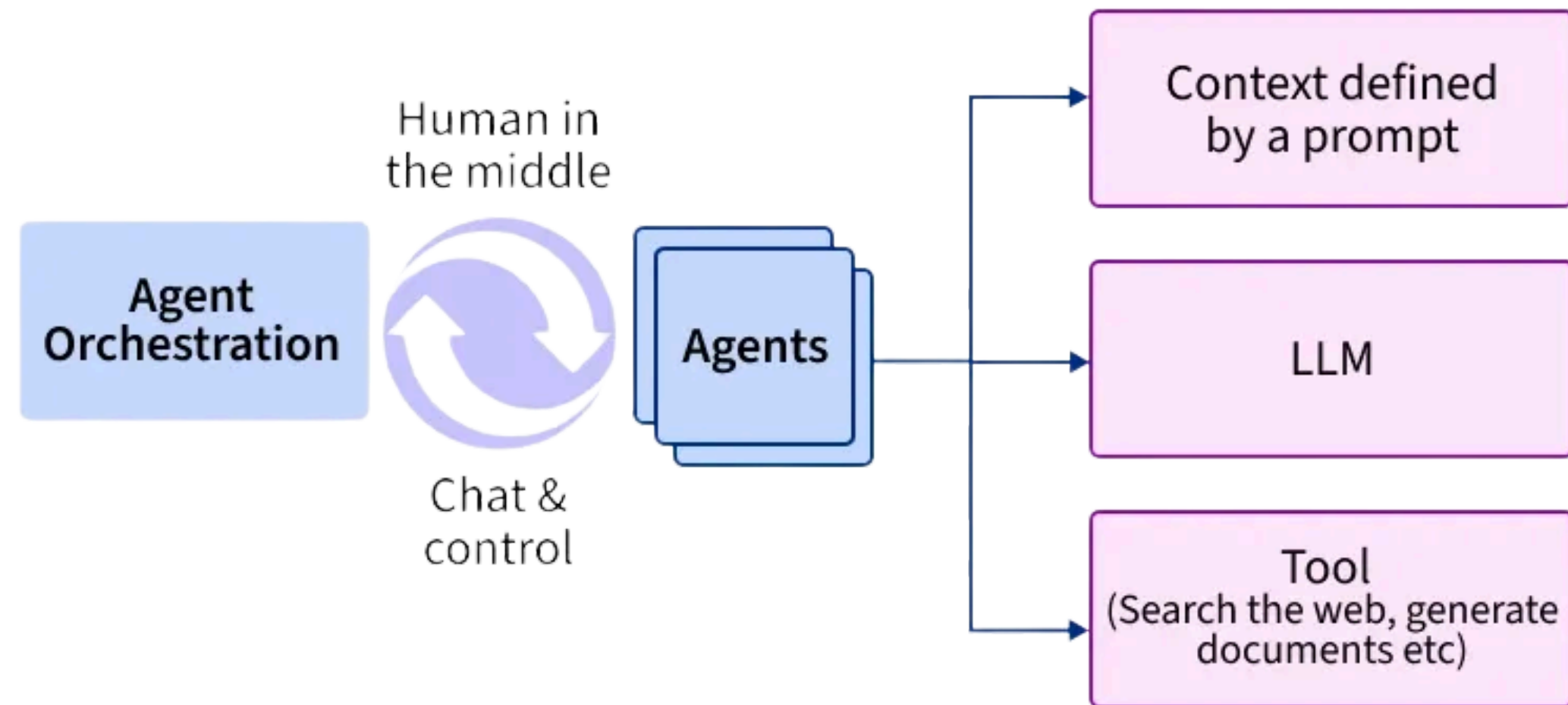
- **Robotics:** Agents can be used to control robots and automate tasks in manufacturing, transportation and other industries.
- **Smart homes and buildings:** They can be used to control heating, lighting and other systems in smart homes and buildings, optimizing energy use and improving comfort.
- **Healthcare:** They can be used to monitor patients, provide personalized treatment plans and optimize healthcare resource allocation.
- **Finance:** They can be used for automated trading, fraud detection and risk management in the financial industry.
- **Games:** They can be used to create intelligent opponents in games and simulations, providing a more challenging and realistic experience for players.



A.I AGENT FRAMEWORK

AI Agent Frameworks

- AI agents have become the backbone of complex task automation, decision-making and workflow management across industries.
- These frameworks helps developers and businesses to build intelligent agents that operate autonomously, coordinate tasks and integrate with a wide range of tools and platforms.
- In this article we will see different AI Agent Frameworks to make our own agents.



1. LangGraph

- LangGraph is designed for building and managing advanced autonomous agents capable of handling independent workflows.
- Its architecture is inspired by directed acyclic graphs (DAGs) which helps in the execution of complex, branching processes such as starting a chatbot session, continuing the conversation and ending it upon achieving goals.

1. LangGraph

Features:

- Smaller Workflow Control: Allows developers to design agent flows at small steps, supporting multi-agent, hierarchical and sequential processes.
- Statefulness: Lets agents remember details from past steps so they can pick up where they left off which also makes it easier to trace and fix problems in their workflows.
- Adaptive Retrieval-Augmented Generation (RAG): It supports advanced knowledge-augmented reasoning, letting agents execute adaptive tasks with minimal human input.
- Integration Ready: Easily connects with various LLMs, vector databases, monitoring, debugging and human-in-the-loop controls.

Use Case: Automating support ticket where agent receives input, gathers context and autonomously routes and resolves queries, escalating only when necessary.

Other Frameworks

- 2. Autogen:** Microsoft-backed multi-agent framework designed for enabling rich, dynamic conversations and automation workflows among AI agents, humans and tools.
- 3. CrewAI:** an open-source framework that lets developers create and manage teams of specialized AI agents.
- 4. Semantic Kernel:** an open-source, lightweight Microsoft SDK that lets developers add advanced AI features such as natural language understanding and generation, to applications written in C#, Python or Java.
- 5. LangChain:** an open-source framework designed to streamline the development of applications powered by large language models (LLMs).

6. No-Code and Visual AI Agent Platforms

For business users, domain experts or fast prototyping, visual and no-code AI agent builders provide drag-and-drop simplicity.

1. **n8n: Visual Agent Development Platform:**

- a. **Workflow Automation Plus AI:** n8n Enables users to construct AI workflows integrating 400+ apps like Notion, Stripe, Google Sheets, etc with LLMs and toolchains all via a graphical interface.
- b. **Trigger-Based Flows:** Agents can be initiated by webhook, chat message or custom event and use built-in nodes for data handling, communication and logic.
- c. **Sales, Marketing, Support:** Automate CRM updates, lead scoring, schedule communications or orchestrate sales pipelines without code.

IT Ops can

⚡ On-board new employees

Sec Ops can

⚡ Enrich security incident tickets

Dev Ops can

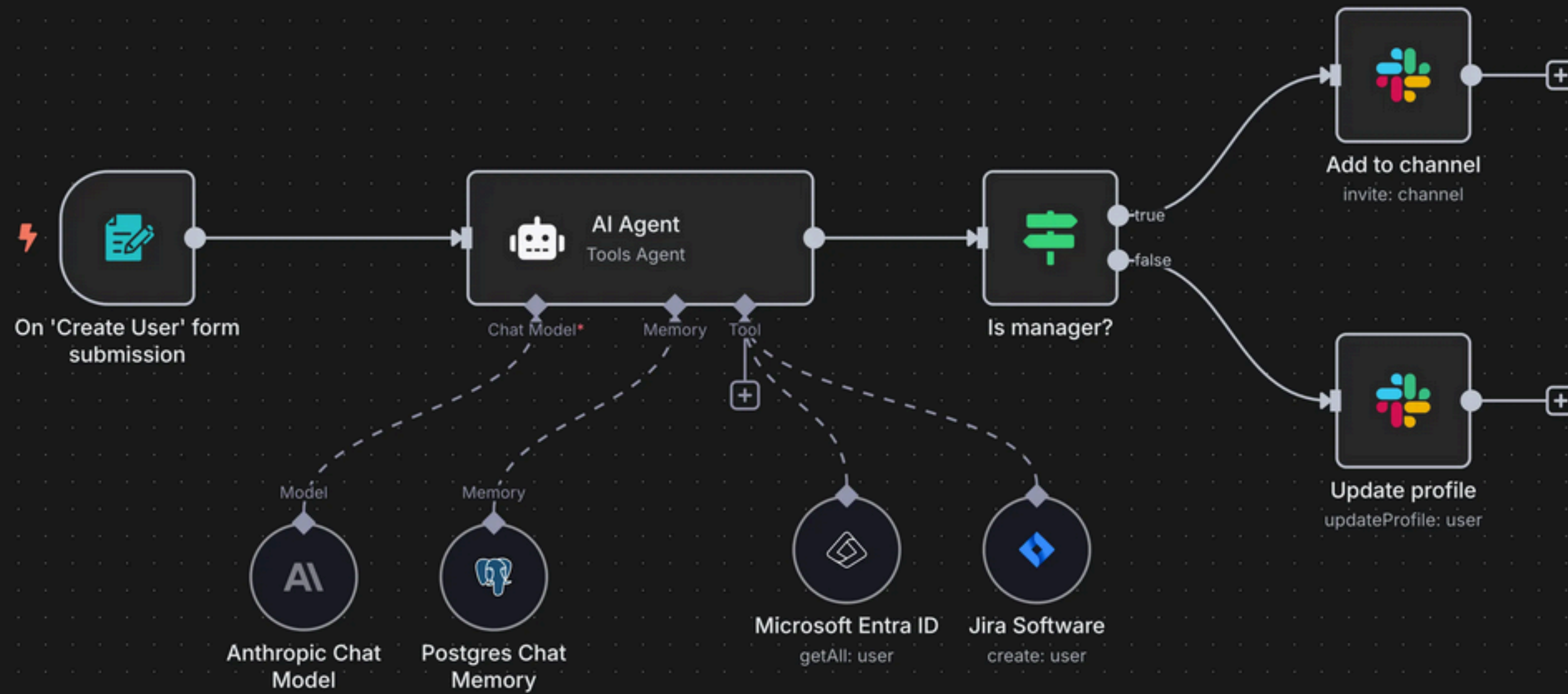
⚡ Convert natural language into API calls

Sales can

⚡ Generate customer insights from reviews

You can

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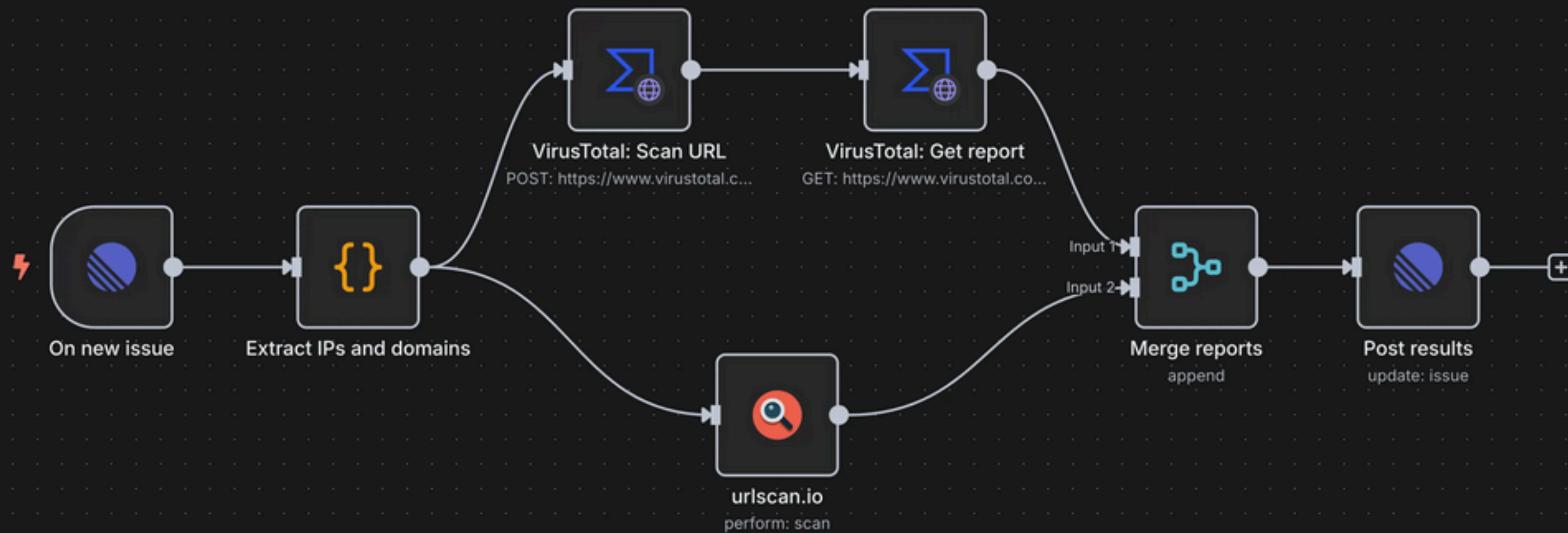
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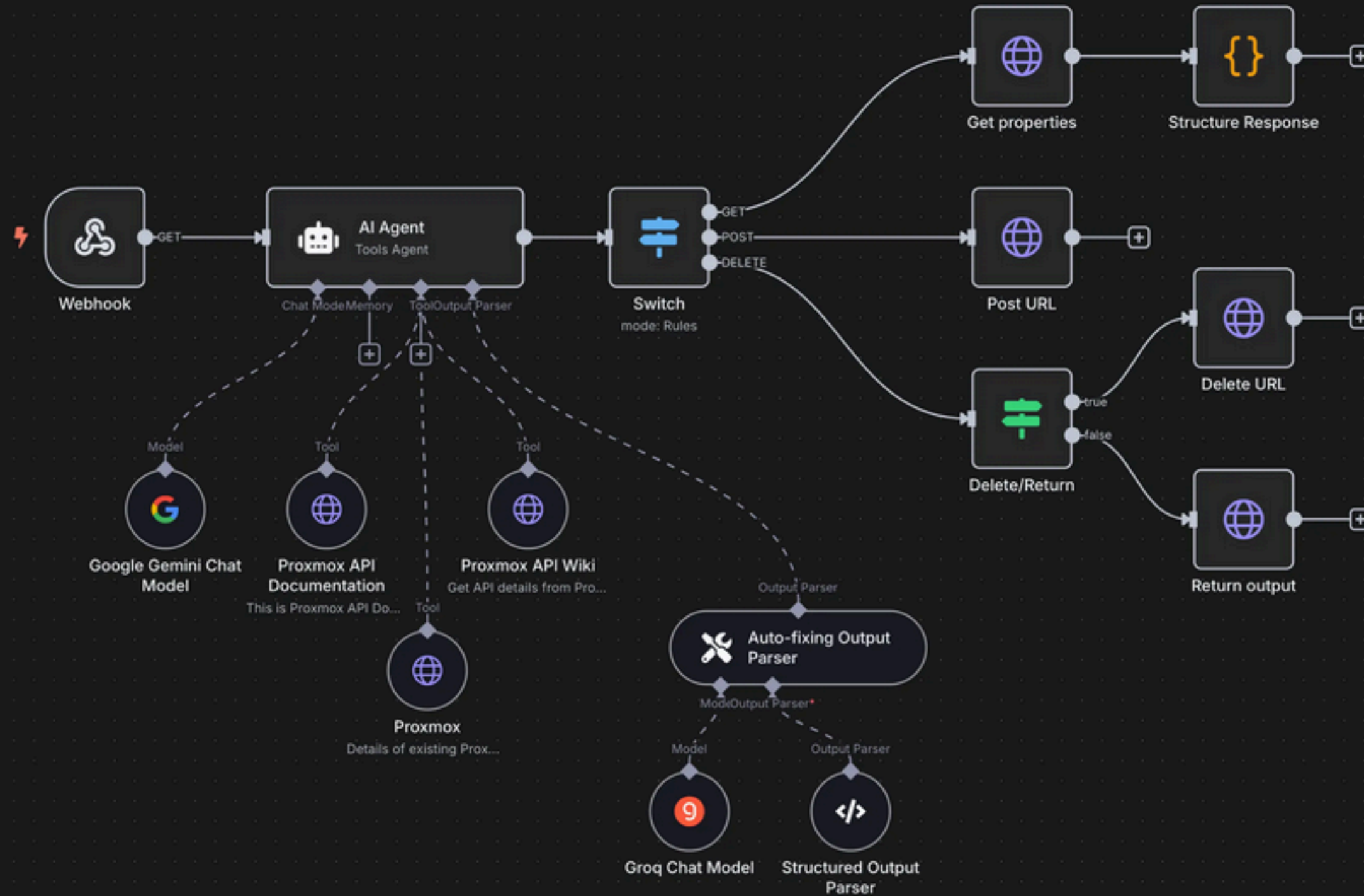
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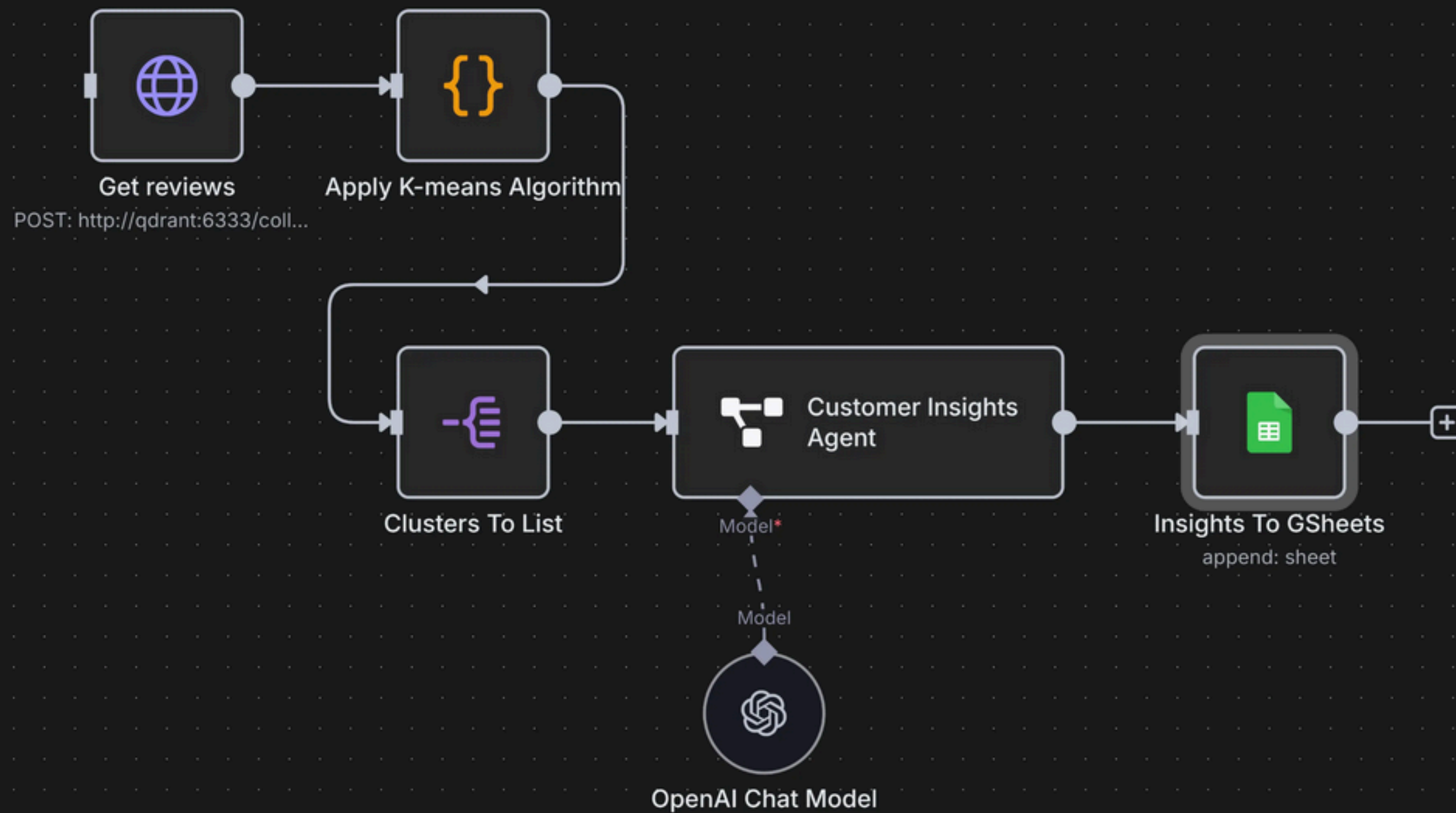
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6. No-Code and Visual AI Agent Platforms

2. Langflow: Modular Drag-and-Drop Framework

- a. **Multi-Agent and RAG Support:** Designed for visually constructing complex agentic workflows including collaborative multi-agent scenarios and RAG tasks.
- b. **LLM and Vector Store Agnostic:** Choose any mainstream or open-source model or vector DB and extend features as needed.
- c. **Prototyping and Collaboration:** Rapidly test, iterate and deploy flows and share, embed or serve flows via APIs for enterprise integration.

Some More Agents Usecases

Coordinate multiple specialized agents (e.g., research, writing, QA) to complete complex workflows.

Perform multi-step research with API access and memory to surface clear, structured takeaways.

Retrieve real-time info from docs, wikis, or internal data to generate up-to-date, verified content.

Break down processes into smaller steps. Decide what happens next – and which agent should execute.



⚡ Multi-Agent Systems: For tasks that need a team

Coordinate multiple specialized agents (e.g., research, writing, QA) to complete complex workflows.

⚡ Deep Research Agents: For mining large amounts of data

Perform multi-step research with API access and memory to surface clear, structured takeaways.

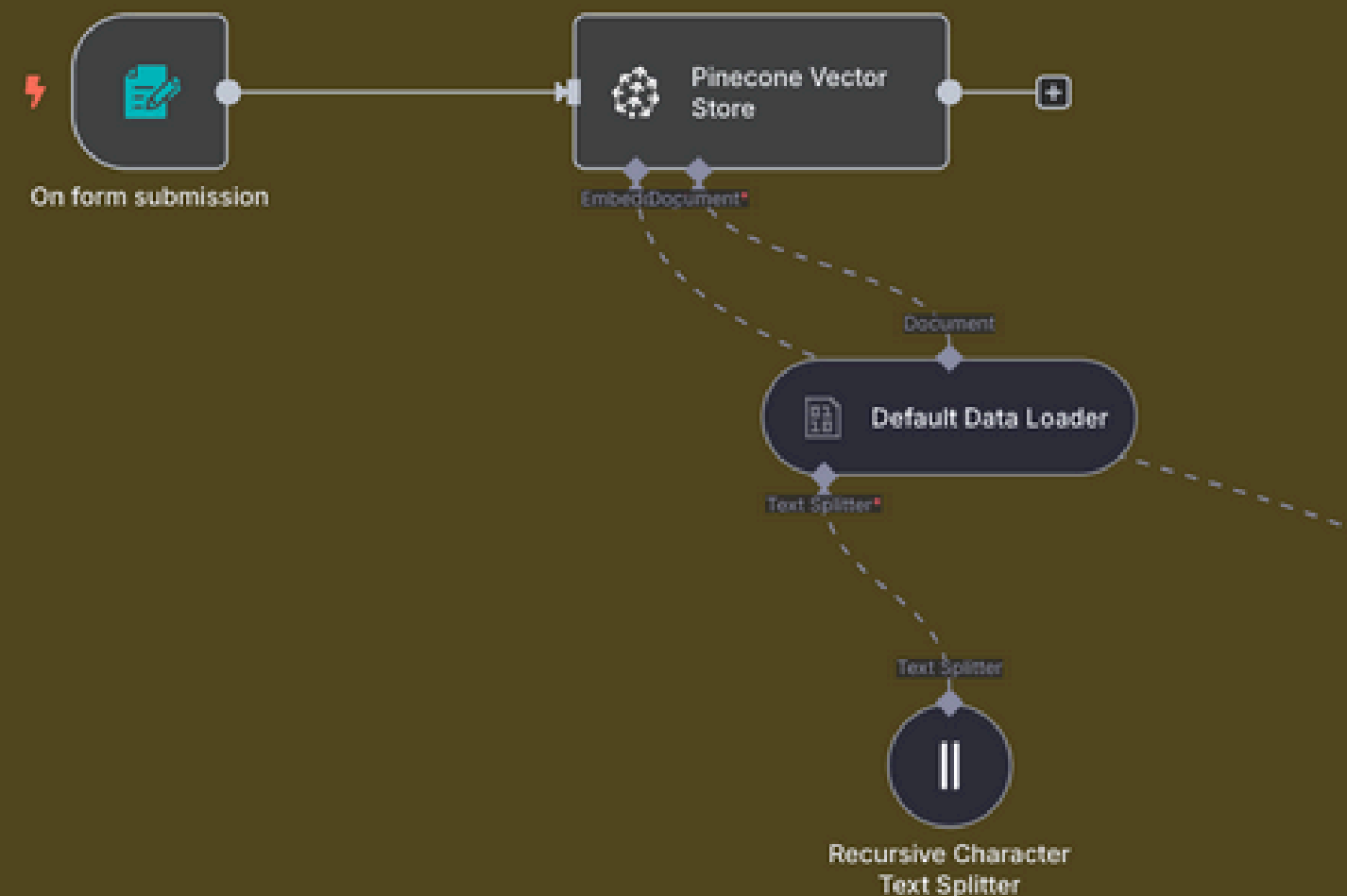
⚡ RAG Agents: For when accuracy absolutely matters

Retrieve real-time info from docs, wikis, or internal data to generate up-to-date, verified content.

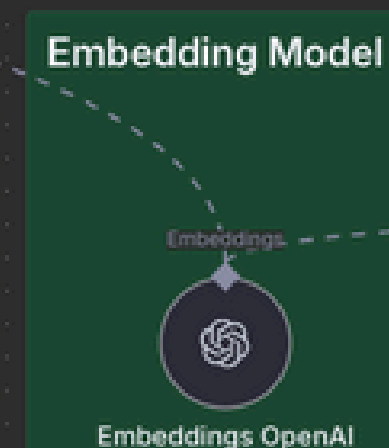
⚡ Planning Agents: For tasks too large to tackle straight on

Break down processes into smaller steps. Decide what happens next – and which agent should execute.

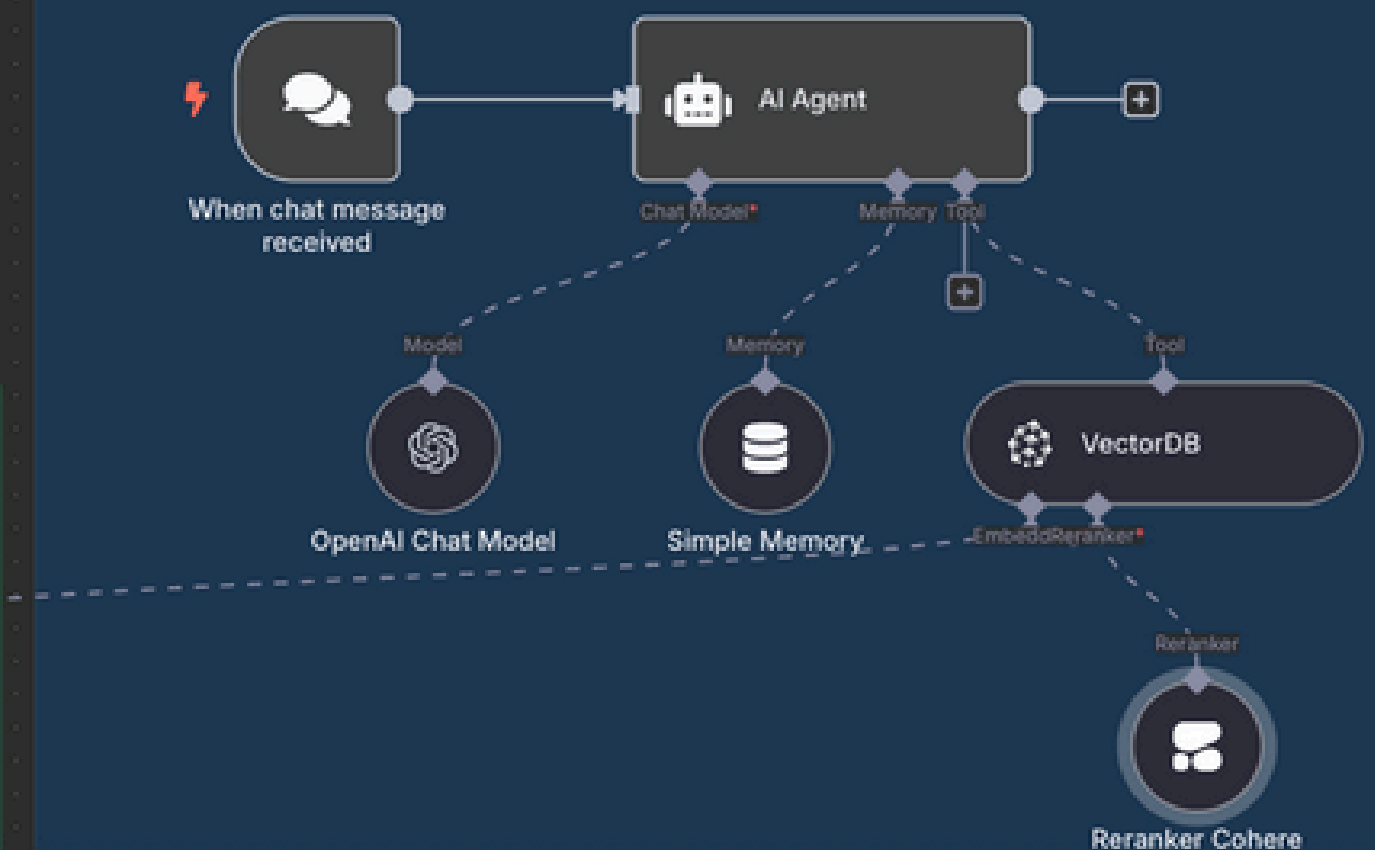
Insert Data to Pinecone



Embedding Model



Chat AI Agent



That's all